```
clc;
clear;
close all;
```

# INPUTS

```
b = [0.0048 0.0193 0.0289 0.0193 0.0048];
a = [1 -2.3695 2.3140 -1.0547 0.1874];

Bc = 8;
Bp = 16;
N = 200;

x = [1 zeros(1,N-1)];
n = 0:N-1;
```

# 1. COEFFICIENT QUANTIZATION

```
S = 2^(Bc-1)-1;

bq = round(b*S)/S;
aq = round(a*S)/S;

% Keep the leading denominator coefficient equal to one
aq = aq/aq(1);
```

# 2. IDEAL AND QUANTIZED RESPONSES

```
y_ideal = filter(b,a,x);
y_coeff = filter(bq,aq,x);

e_coeff = y_ideal-y_coeff;

MaxCoeffError = max(abs(e_coeff));

SQNR_coeff = 10*log10( ...
    sum(y_ideal.^2)/(sum(e_coeff.^2)+eps));
```

# 3. PRODUCT QUANTIZATION

```
Sp = 2^(Bp-1)-1;

y_prod = zeros(1,N);
state = zeros(1,4);

for k = 1:N

    feedforward = bq*x(k);
    feedback = -aq(2:5).*state;
```

```
    products = [feedforward feedback];

    products_q = round(products*Sp)/Sp;

    accumulator = sum(products_q);

    % Quantize the output
    y_prod(k) = round(accumulator*Sp)/Sp;

    % Update delay states
    state = [y_prod(k) state(1:3)];

end

e_prod = y_coeff-y_prod;

MaxProdError = max(abs(e_prod));

SQNR_prod = 10*log10( ...
    sum(y_coeff.^2)/(sum(e_prod.^2)+eps));
```

# 4. ZERO-INPUT LIMIT-CYCLE TEST

```
state = [0.01 0.01 0.01 0.01];
y_zero = zeros(1,N);

for k = 1:N

    acc = -aq(2:5)*state.';

    y_zero(k) = round(acc*Sp)/Sp;

    state = [y_zero(k) state(1:3)];

end

LimitAmplitude = max(abs(y_zero));
LimitSamples = find(abs(y_zero)>1e-12);
```

# 5. OVERFLOW AND SATURATION TEST

```
xo = 1.2*ones(1,N);

y_over = filter(bq,aq,xo);

y_sat = min(max(y_over,-1),1);

OverflowCount = sum(abs(y_over)>1);

MaxOverflow = max(abs(y_over));
```

# 6. DISPLAY RESULTS

```
fprintf('\n=======================================\n');
fprintf(' IIR FINITE WORD-LENGTH RESULTS\n');
fprintf('=======================================\n');

fprintf('Coefficient/output error = %.6e\n',MaxCoeffError);
fprintf('Coefficient SQNR = %.2f dB\n',SQNR_coeff);

fprintf('Product quantization error = %.6e\n',MaxProdError);
fprintf('Product SQNR = %.2f dB\n',SQNR_prod);

fprintf('Zero-input maximum amplitude = %.6e\n',LimitAmplitude);
fprintf('Number of nonzero zero-input samples = %d\n',length(LimitSamples));

fprintf('Overflow samples = %d\n',OverflowCount);
fprintf('Maximum absolute output = %.6e\n',MaxOverflow);
```

```
=======================================
 IIR FINITE WORD-LENGTH RESULTS
=======================================
Coefficient/output error = 4.405764e-03
Coefficient SQNR = 32.23 dB
Product quantization error = 1.651951e-01
Product SQNR = 1.57 dB
Zero-input maximum amplitude = 9.216590e-03
Number of nonzero zero-input samples = 177
Overflow samples = 194
Maximum absolute output = 1.359910e+00
```

# 7. RESULTS TABLE

```
Test = {
    'Coefficient Quantization'
    'Product Quantization'
    'Zero-Input Test'
    'Overflow Test'
};

MaximumValue = [
    MaxCoeffError
    MaxProdError
    LimitAmplitude
    MaxOverflow
];

SQNR_dB = [
    SQNR_coeff
    SQNR_prod
    NaN
    NaN
];
```

```
T = table(Test,MaximumValue,SQNR_dB);

fprintf('\nRESULTS TABLE\n');
disp(T);
```

```
RESULTS TABLE
                Test                 MaximumValue    SQNR_dB
    _____________________________    ____________    _______

    {'Coefficient Quantization'}      0.0044058      32.233
    {'Product Quantization'    }         0.1652      1.5673
    {'Zero-Input Test'         }      0.0092166         NaN
    {'Overflow Test'           }         1.3599         NaN
```

# 8. COEFFICIENT DISPLAY

```
fprintf('\nOriginal numerator coefficients:\n');
disp(b);

fprintf('Quantized numerator coefficients:\n');
disp(bq);

fprintf('Original denominator coefficients:\n');
disp(a);

fprintf('Quantized denominator coefficients:\n');
disp(aq);
```

```
Original numerator coefficients:
    0.0048    0.0193    0.0289    0.0193    0.0048

Quantized numerator coefficients:
    0.0079    0.0157    0.0315    0.0157    0.0079

Original denominator coefficients:
    1.0000   -2.3695    2.3140   -1.0547    0.1874

Quantized denominator coefficients:
    1.0000   -2.3701    2.3150   -1.0551    0.1890
```

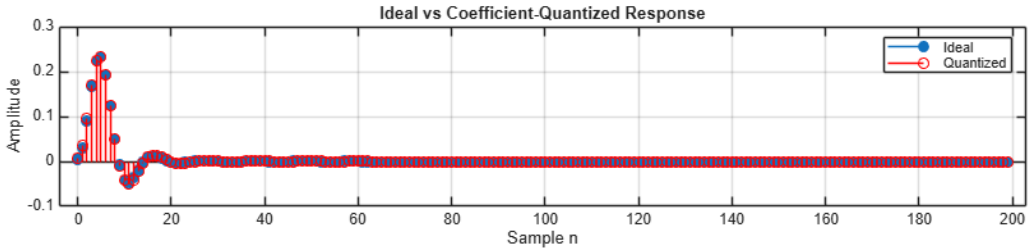
# 9. GRAPH 1: IDEAL VS QUANTIZED RESPONSE

```
figure;

stem(n,y_ideal,'filled');
hold on;
```

```
stem(n,y_coeff,'r');

grid on;
xlabel('Sample n');
ylabel('Amplitude');
title('Ideal vs Coefficient-Quantized Response');
legend('Ideal','Quantized');
```

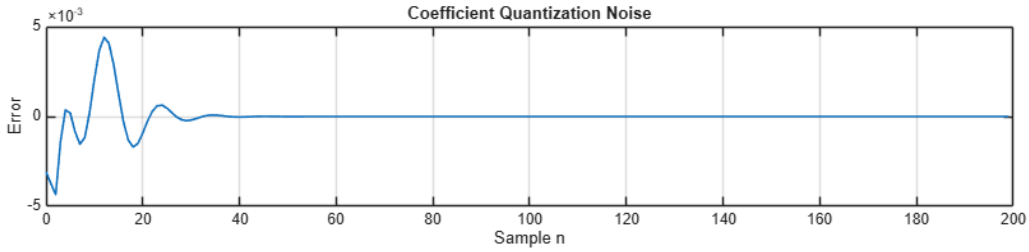



# 10. GRAPH 2: COEFFICIENT QUANTIZATION ERROR

```
figure;

plot(n,e_coeff,'LineWidth',1.2);

grid on;
xlabel('Sample n');
ylabel('Error');
title('Coefficient Quantization Noise');
```

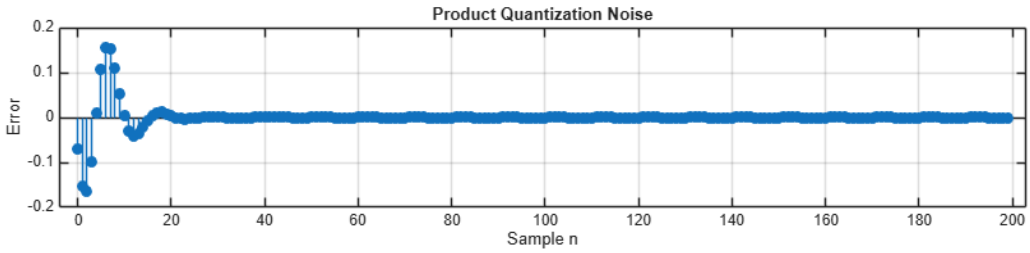



# 11. GRAPH 3: PRODUCT QUANTIZATION ERROR

```
figure;

stem(n,e_prod,'filled');

grid on;
xlabel('Sample n');
ylabel('Error');
title('Product Quantization Noise');
```

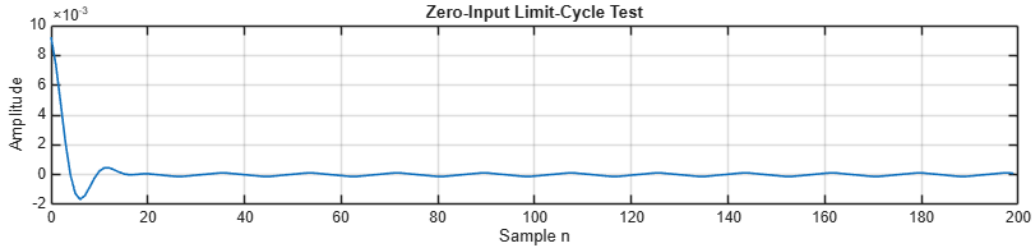



# 12. GRAPH 4: ZERO-INPUT RESPONSE

```
figure;

plot(n,y_zero,'LineWidth',1.2);

grid on;
xlabel('Sample n');
ylabel('Amplitude');
title('Zero-Input Limit-Cycle Test');
```

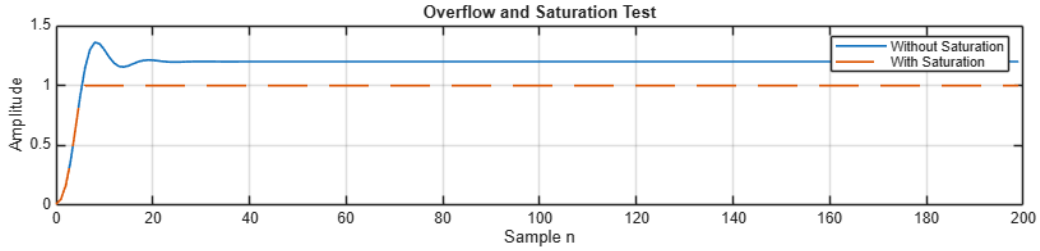



# 13. GRAPH 5: OVERFLOW AND SATURATION

```
figure;

plot(n,y_over,'LineWidth',1.2);
hold on;
plot(n,y_sat,'--','LineWidth',1.2);

grid on;
xlabel('Sample n');
ylabel('Amplitude');
title('Overflow and Saturation Test');
legend('Without Saturation','With Saturation');
```



# 14. GRAPH 6: IDEAL POLE-ZERO PLOT

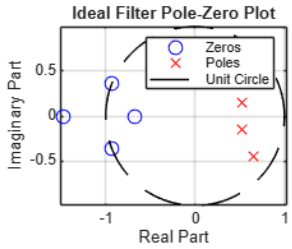
```
figure;

z = roots(b);
p = roots(a);

plot(real(z),imag(z),'ob','MarkerSize',8);
hold on;
plot(real(p),imag(p),'xr','MarkerSize',8);

theta = linspace(0,2*pi,400);
plot(cos(theta),sin(theta),'k--');

axis equal;
grid on;
xlabel('Real Part');
ylabel('Imaginary Part');
title('Ideal Filter Pole-Zero Plot');
legend('Zeros','Poles','Unit Circle');
```



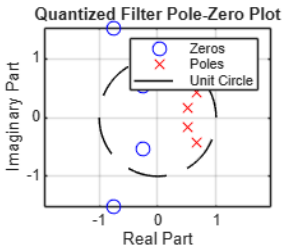

# 15. GRAPH 7: QUANTIZED POLE-ZERO PLOT

```
figure;

zq = roots(bq);
pq = roots(aq);

plot(real(zq),imag(zq),'ob','MarkerSize',8);
hold on;
plot(real(pq),imag(pq),'xr','MarkerSize',8);
plot(cos(theta),sin(theta),'k--');

axis equal;
grid on;
xlabel('Real Part');
ylabel('Imaginary Part');
title('Quantized Filter Pole-Zero Plot');
legend('Zeros','Poles','Unit Circle');
```




# FINAL RESULT

```
fprintf('\n=======================================\n');
fprintf(' FINAL RESULT\n');
fprintf('=======================================\n');

fprintf('1. Coefficient quantization completed.\n');
fprintf('2. Product quantization completed.\n');
fprintf('3. Zero-input response evaluated.\n');
fprintf('4. Overflow and saturation evaluated.\n');
fprintf('5. Seven graphs generated.\n');
```

```
=======================================
 FINAL RESULT
=======================================
1. Coefficient quantization completed.
2. Product quantization completed.
3. Zero-input response evaluated.
4. Overflow and saturation evaluated.
5. Seven graphs generated.
```

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